Managerial Coordination in Decentralized Markets

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Abstract

Why do managers coordinate resource allocation in market economies? We develop a repeated-game model showing that managerial coordination can reduce the cost of incentivizing performance when individual demands are volatile and needs are specialized. This benefit, however, must be balanced against the incentive rents required by the managers themselves. The trade-off leads to the endogenous sorting of market participants into managerial coordination and decentralized exchange. We generate testable predictions for the drivers of managerial coordination. We characterize the impacts of managerial coordination on economic efficiency, the division of labor, matching market flows, and rent distribution. We discuss applications to professional services, global sourcing, and online platforms.

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1 Introduction

A fundamental puzzle in economic theory is the prevalence of managerial coordination. In principle, buyers and producers in many markets could transact directly with one another. Yet exchange is frequently mediated by centralized intermediaries that coordinate allocations, monitor performance, and enforce relational contracts (Alchian and Demsetz 1972). The theory of the firm has made considerable progress in modeling such manager-monitors, but much of this work has focused on understanding individual relational contracts (e.g., Baker, Gibbons and Murphy 1994, 2002; Levin 2003; Malcomson 2013; Gil and Zanarone 2017). Recent work extends these frameworks to decentralized matching markets, illuminating how relational contracts arise when agents meet and contract freely (Board and Meyer-ter-Vehn 2015; Fahn and Murooka 2025). Despite these advances, a key gap remains: the literature lacks formal models that explain why and when centralized monitors add value by coordinating allocations in otherwise decentralized relational contracting markets.

This paper develops a tractable repeated-game model of managerial coordination in decentralized matching markets. The core of our analysis is a novel economic mechanism: we show that, in settings characterized by moral hazard and demand fluctuations, managers can reduce the cost of motivating performance by coordinating the assignment of buyers to producers. The model allows us to analyze the endogenous sorting of agents into managerial coordination and decentralized exchange. We also characterize the equilibrium effects of managerial coordination on output, the division of labor, market flows, and rent sharing. We then apply the framework as a lens to interpret recent social and economic transformations. Our work extends the literature on intermediaries (e.g., Rubinstein and Wolinsky 1987; Biglaiser 1993; Spulber 1999) by showing how managerial coordination can arise even in the absence of search frictions or adverse selection.

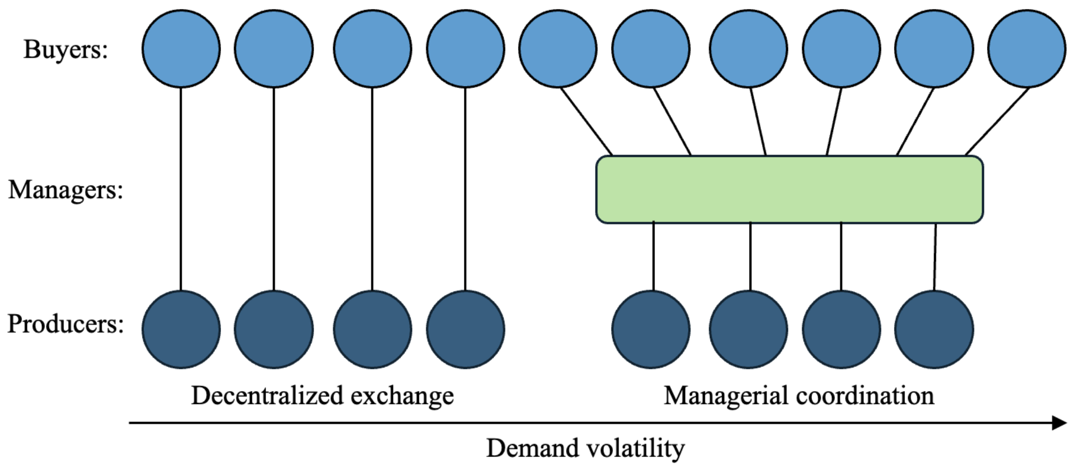
Our framework is a variant of MacLeod and Malcomson (1989). Following previous models of environments with imperfect contractual enforcement, we assume that buyers cannot compel producers to exert effort using formal contracts, so buyers must motivate producers using future surplus in relational contracts. Buyers and producers meet and enter contracts in frictionless matching markets without search delays or hidden payoff-relevant information, as in Board and Meyer-ter-Vehn (2015). Departing from prior models, we introduce a new feature—fluctuating buyer demand—to capture a realistic and relevant aspect of repeated business interactions (Macchiavello and Morjaria 2015). We also introduce managers who cannot themselves produce, but can manage—i.e., monitor and allocate—a large number of producers. In equilibrium, managers endogenously enter multiple relational contracts on both sides of the market, aggregate fluctuating demand, coordinate assignments, and keep track of producer performance.

The model reveals that managerial coordination involves a trade-off between demand smoothing and double marginalization. The benefit of managers is that they aggregate fluctuating buyer demand, coordinate assignments between buyers and producers, and thereby prevent producers from becoming idle or unmatched. Therefore, managers can promise longer and more productive relational contracts, thereby relaxing dynamic enforcement constraints and reducing the incentive rent needed to motivate producers to exert effort. However, managers must themselves be motivated by an additional incentive rent. The cost of managers is thus a form of double marginalization.

Because of this trade-off, it is found that buyers sort into managerial coordination and decentralized exchange. Their optimal choice depends on individual characteristics, such as their individual demand volatility. It also depends on market-level characteristics, such as market tightness. If the matching market is sufficiently tight, managers emerge as centralized contracting nexuses. Buyers with short-lived demand contract with managers. Buyers with long-lived demand contract directly with producers in a sea of decentralized pairings. Figure 1 illustrates the relational contracting networks that endogenously emerge in the unique steady-state equilibrium of the exchange economy.

Several comparative statics are derived regarding the determinants and impacts of managerial coordination. First, when compared to managed producers, directly contracted producers have higher average incentive pay, more dispersed pay, higher separation rates, and higher idleness. Second, the introduction of managers into an economy benefits buyers

Figure 1: Endogenous relational contracting networks in steady-state equilibrium



with fluctuating demand, but reduces the pay of producers initially earning pay premiums. As such, managerial coordination both increases and redistributes total surplus.

In extensions, we analyze the relationships between managerial coordination, specialization, and public observability of manager performance. We show that managed producers are more specialized and that managerial coordination increases producer specialization. Moreover, buyers are more likely to choose managerial coordination if there are gains from producer specialization. We also show that, unlike intermediation that mitigates search frictions or adverse selection, managerial coordination may *increase* as manager performance becomes publicly observed. The reason is that public observability allows buyers to better coordinate on collective punishments against managers who shirk.

Several applications illustrate how our framework can be used to analyze organizational forms that increasingly displace traditional decentralized exchange. Labor markets, for example, have been fundamentally altered by the rise of professional service firms that monitor and deploy large numbers of workers (e.g., legal, accounting, IT, and HR firms).A parallel revolution has reshaped international trade, where global sourcing firms (e.g., Li & Fung and Shein) increasingly manage vast supplier networks and coordinate significant shares of world production. More recently, consumer markets have been radically restructured by digital platforms like Uber and Airbnb, which allocate and monitor (i.e., manage) increasingly large numbers of independent contractors. Connections between our model and the empirical literature on professional service firms are discussed throughout the paper. As we shall show, the model generates realistic predictions not only for the drivers of professional service outsourcing, but also for the effects of professional service firms on both workers and the labor market as a whole.

The rest of the paper proceeds as follows. Section 2 discusses related literature. Section 3 introduces our model and empirical applications. Section 4 characterizes and compares direct and managed contracts, taking matching market conditions as given. Section 5 derives the steady-state market equilibrium. Section 6 pursues extensions. Section 7 concludes.

2 Related Literature

Our work contributes to a growing literature on repeated-game models of relational contracts in decentralized matching markets (Shapiro and Stiglitz 1984; MacLeod and Malcomson 1989, 1998; Yang 2008; Board and Meyer-ter-Vehn 2015; Fahn 2017; Powell 2019; Li 2022; Fahn and Murooka 2025). MacLeod and Malcomson (1989) lays the game-theoretic foundation for our modeling approach, while Board and Meyer-ter-Vehn (2015) provides a crisp treatment that we build upon. To these models, we introduce fluctuating demand and managers with neither intrinsic demand nor productive capability. The managers aggregate fluctuating demand and coordinate assignments, easing a tension between incentive provision and flexible allocations, and thereby enabling trade and specialization.

Several works studying multilateral relational contracts in fluctuating environments are closely related. Board (2011) showed that the tension between incentive provision and flexible allocations can lead buyers to restrict their number of trading partners. Andrews and Barron (2016) showed that it can cause dynamic allocations to depend on payoff-irrelevant past performance. Li and Powell (2020) highlighted that a similar tension can lead agents to interact in multiple activities. This paper is, however, the first to show how the same tension leads to centralized managerial coordination.[1]

The model developed here is closely related to informal intuitions proposed in early contract-based theories of the firm. Alchian and Demsetz (1972) proposed that the essence of the firm is the presence of a central contracting party that monitors performance and excludes underperforming team members from future participation. Jensen and Meckling (1976, pp. 310–311) expanded on this view and suggested that "contractual relations are the essence of the firm, not only with employees but with suppliers, customers, creditors, and so on." Demsetz (1988, pp. 154) concurred that "the firm properly viewed is a 'nexus' of contracts" and elaborated that the key attributes of firm-like organization are continuity of association, specialization, and managerial direction. Our model clarifies why and when centralized contracting intermediaries coordinate resource allocations.

Our results can be viewed as providing a novel micro-founded answer to a variant of the question first posed by Coase (1937): Why do centralized resource allocators emerge in decentralized markets? We find that centralized allocators can better motivate performance in relational contracting markets when demand is volatile and needs are specialized. Importantly, they do so even in the absence of search, bargaining, and contract-writing costs, as well as hidden information about agent types.

Similar ideas to ours can be found in the literatures on supply chain management and market institutions. Belavina and Girotra (2012) developed a model of relational intermediaries in supply chains with two buyers, two sellers, and non-transferable utilities. Our transferable utility model is similar but enables a fuller characterization of the determinants and welfare consequences of this type of intermediation. Milgrom, North and Weingast (1990) used a model of repeated Prisoner's Dilemma games to show that private judges in medieval trade can create trust with less observability than reputation systems.[2] This paper follows a similar logic. However, our formal approach is different, and our focus is not the historical role of legal and regulatory institutions. In our view, the problem of trust remains

[1]For surveys of recent work on relational contracts in economics, see MacLeod (2007), Malcomson (2013), Gil and Zanarone (2017), Macchiavello and Morjaria (2023), and Muehlheusser, Fahn and MacLeod (2023).

[2]Related papers on market institutions include Greif (1993, 2006), Greif, Milgrom and Weingast (1994), Calvert (1995), and Fafchamps (2004).

pervasive despite the development of modern legal and regulatory institutions. The need for trust explains the ubiquity of managerial coordination.

Our work is quite different from the existing formal literature on firm theory, however. We do not attempt to define what a firm is (cf. Grossman and Hart 1986), nor do we analyze how contracts within firms differ from those between firms (cf. Baker, Gibbons and Murphy 2002).[3] Rather, our focus is to show why and when market participants choose managerial coordination over decentralized exchange in relational contracting environments. Our approach follows the spirit of Cheung (1983), who suggested that it may be "futile to press the issue of what is or is not a firm," since "a 'firm' may be as small as a contractual relationship between two input owners or, if the chain of contracts is allowed to spread, as big as the whole economy." Our contribution characterizes how contractual networks form in environments with imperfect enforcement. The identified trade-offs are then used to understand firm-like organizations, such as professional service firms.

Our contribution adds to a literature on general equilibrium models that feature managers. Lucas (1978) provided a model of occupational choice in which managerial talent is scarce and highly complementary to production workers, leading to high pay for managers. Garicano (2000) pioneered a model of knowledge hierarchies, in which managers function as expert problem-solvers who help less skilled workers. Similar to this literature, our model features predictions regarding organizational structure and their equilibrium distributional impacts. Existing approaches, however, abstract from the incentive problems that many scholars believe managers exist to remedy. The model of managers presented here is the first to generate predictions regarding efficiency, inequality, and organizational structure with incentive-based microeconomic foundations.

The broader literature on intermediaries features many related but fundamentally different

[3]Beginning with Coase (1937) and Williamson (1971, 1975, 1985), many forces have been cataloged as determinants of firm boundaries. A prominent strand of the formal literature studies how firm boundaries are shaped by the optimal allocation of asset ownership (Grossman and Hart 1986; Baker, Gibbons and Murphy 2002; Gibbons 2005). Another explores the differences between service and employment contracts, emphasizing differences in bargaining and contract-writing costs (Wernerfelt 1997, 2015, 2016; Hart and Moore 2008; Levin and Tadelis 2010; Tadelis and Williamson 2013), or differences between spot and relational contracting (Simon 1951; Bolton and Rajan 2003; Raith 2022).

ideas. To date, many models have been devised to study middlemen who overcome search frictions (e.g. Rubinstein and Wolinsky 1987). Another set of models studies intermediaries who overcome adverse selection (e.g. Biglaiser 1993).[4] However, it is a puzzle for these theories why the advent of the Internet has not led to radical disintermediation, but has instead spurred the growth of intermediary firms (Belavina and Girotra 2012; Bergeaud et al. Forthcoming). Here we formalize a novel mechanism where intermediaries overcome moral hazard through coordinating buyer-producer assignments. This flavor of intermediation has rarely been formally studied, but is relevant for understanding the role of managers and firm-like organizations in the economy. We show that this form of intermediation may grow as information becomes more abundant.

3 Model

Basics. Time is discrete and infinite, $\tau \in \{0, 1, ...\}$. Buyers are heterogeneous, indexed by i, and have unit mass. Producers are identical and have measure n. Managers are identical and have a finite number K. All players are infinitely-lived. They have a common discount factor $\delta \in (0, 1)$. The measure of producers, n, is determined by endogenous entry subject to entry cost $C > 0$. Entry cost C represents training or opportunity cost.

Demand realization. At the start of each period, the demand of each buyer i, denoted as $d_{i,\tau} \in \{0, 1\}$, is realized and publicly observed.[5] Each buyer's demand $d_{i,\tau}$ is independently drawn following a Markov process. The demand-switching probabilities are determined by each buyer's publicly known type $\alpha_i = (\alpha_{1i}, \alpha_{0i})$: with probability α_{1i}, buyer i's demand switches from 1 to 0; with probability α_{0i}, buyer i's demand switches back from 0 to 1. The

[4] Recent models of relational contracting intermediaries emphasize very different mechanisms. Fong and Li (2017) shows that relational contracts can be deepened by the presence of a non-strategic supervisor carrying out subjective performance reviews. Troya-Martinez and Wren-Lewis (2023) explores a model where a manager may receive kick-backs and show that such managers can improve relational contracts in environments with commitment difficulty.

[5] To focus on the relational incentives, we abstract from any adverse selection problem in the model. Therefore, when contracting with buyers, producers know whom they are dealing with and have the correct expectation of how the relationships will go.

buyers types have distribution F on $[0, 1] \times [0, 1]$.

Matching. After demand is realized, matched players may choose to separate. Buyers may then open vacancies for either producers or managers. Since managers cannot produce by themselves, they fulfill buyer demand by matching with producers in turn.

The timing is as follows: (1) buyers open either manager vacancies or producer vacancies, (2) managers are matched with manager vacancies, (3) managers open producer vacancies, and (4) unmatched producers are matched with producer vacancies. Each buyer may open at most one vacancy, while managers may match with a continuum of buyers and producers.

Following Board and Meyer-ter-Vehn (2015), the matching of producer vacancies follows three axioms. First, matching is *frictionless*, meaning that all vacancies are filled subject to the managers' or the producers' participation constraints. Second, matching is *anonymous*, meaning that each producer's probability of becoming matched with a vacancy does not depend on their past behavior. The assumption of anonymous matching for producers is standard and simplifies analysis.[6] Third, matching is *random*, i.e., unmatched producers have the same probability of becoming matched.[7]

The matching of manager vacancies is assumed to be frictionless, random, but *not* anonymous. In other words, if a manager violates its relational contract with a buyer and becomes unmatched, the buyer remembers the manager's violation and refuses to match with the manager forever. This assumption captures the idea that the market for managers is comparatively small.

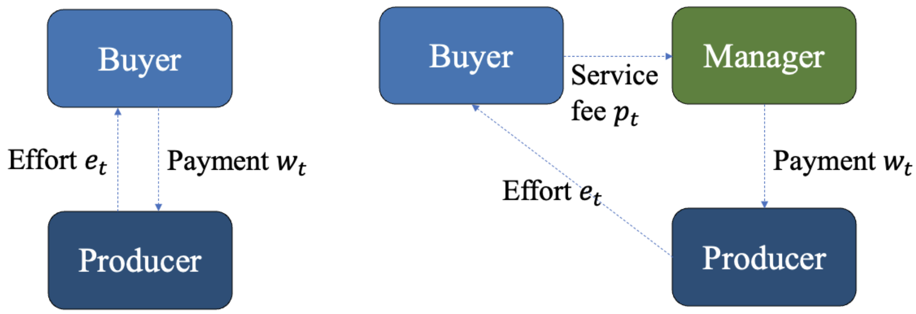
We call a buyer-producer match a *direct match*, since it does not involve a manager. A buyer-manager-producer match is called a *managed match*.

Direct match. Let $t \in \{1, \ldots\}$ denote the period in a match. If a buyer i matches with a producer, the remainder of the period proceeds as follows. First, the buyer makes a payment

[6] See, e.g., Shapiro and Stiglitz (1984); MacLeod and Malcomson (1998); Board and Meyer-ter-Vehn (2015).

[7] Board and Meyer-ter-Vehn (2015) analyzes relational contracts in a frictionless matching market where matched producers may receive on-the-job offers. They show that this leads to heterogeneous productivity across otherwise identical matches. We rule this possibility out for simplicity.

Figure 2: Structure of direct and managed contracts



(a) Direct contract (b) Managed contract

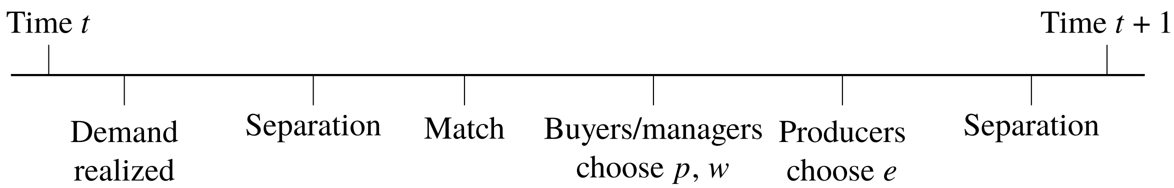
$w_t \geq 0$ to the producer. The producer then chooses an effort level, denoted by $e_t \in \{0, 1\}$. The cost of effort is given by $c(e_t)$, where $c(0) = 0$ and $c(1) = c > 0$. The effort generates an output for the buyer only if demand is positive, so $y_{i,t} = yd_{i,t}e_t$. Effort and output are observable by the buyer but are not verifiable by a court.

Managed match. If a buyer matches with a manager, she pays a service fee $p_t \geq 0$ to the manager. The manager chooses to assign one or none of its producers to the buyer. The manager pays w_t to the producer. The producer then exerts costly effort e_t and produces output y_{it} for the buyer. Effort and output are observable by both the buyer and the manager, but are not verifiable by a court.

Separation. Either party in a match may choose to separate from their match both after demand realization and after production. If separation occurs after demand realization, both parties can participate in the producer or manager market matching in the current period. However, if separation occurs after production, they need to wait till the next period to rematch.

Payoffs. In a direct match, the buyer's payoff is $y_{i,t} - w_t$ and the producer's is $w_t - c(e_t)$ in each period t. In a managed match, the buyer's payoff is $y_{i,t} - p_t$, the manager's payoff is $p_t - w_t$, and the producer's payoff is $w_t - c(e_t)$ in each period t.

Figure 3: Timeline of one period



At the start of the period, buyer demand is realized and publicly observed. Separations may then occur. Unmatched buyers can open manager vacancies and match with managers. Unmatched buyers and managers can then open producer vacancies and match with unmatched producers. In a direct contract, the buyer pays the producer and the producer exerts effort. In a managed contract, the buyer pays the manager, the manager assigns a producer to the buyer and pays the producer, and the producer then exerts effort. Separation may occur at the end of the period.

Remark. This model has two key features that are not present in standard models of relational contracting in frictionless matching markets (e.g., Shapiro and Stiglitz (1984) and MacLeod and Malcomson (1998)). First, we allow buyers to have fluctuating demand. Second, we introduce managers who can enter relational contracts with a multitude of players on both sides of the market. These managers do not have intrinsic demand or productive ability. What they can do, instead, is to match with a large set of buyers and producers, coordinate their matched producers, and monitor the producers' performance. As we will show in Section 4 below, when demand is volatile and the cost of idleness is high, managers can sustain cheaper relational contracts on both sides of the market by reassigning their producers across buyers. Buyers may therefore prefer managerial coordination to direct relational contracting.

3.1 Empirical Applications

This subsection shows how the model can be used to think about professional service firms, global sourcing firms, and online service platforms.

Example 1 (Professional service firms)**.** Consider an entrepreneur (*buyer*) who needs a service performed. She can fulfill the demand either by employing an in-house worker (*producer*) or contracting it out to a professional service firm (*manager*) that allocates its employees to clients. Entrepreneurs may make these employ-or-outsource decisions for various professional services, including cleaning, security, accounting, legal, IT, HR, and temporary help services. These decisions determine the labor boundary of firms. One might, for example, define *employment* as a direct relational contract between an employer and a worker, while *outsourcing* may be defined as a managed contract in which an professional service firm employs workers and assigns them to different entrepreneurs. Under these definitions, the model can be used to predict the determinants and effects of professional service outsourcing.[8]

Example 2 (Global sourcing firms)**.** Consider a retailer (*buyer*) who needs a manufacturing service performed. It can fulfill the demand either by directly contracting with a manufacturer (*producer*) or contracting through a global sourcing firm (*manager*) such as Li & Fung or Shein. Such firms maintain a large number of relationships with upstream manufacturers and function as nexuses of relational contracts (Belavina and Girotra 2012). We can define the service contract between the retailer and the manufacturer as a direct contract and that between the retailer and the sourcing firm as a managed contract. Under these definitions, the model can be used to analyze the scope and effects of global sourcing firms.[9]

Example 3 (Online service platforms)**.** Consider a consumer (*buyer*) who needs a service performed. It can fulfill the demand either by directly contracting with a service provider (*producer*) or contracting through an online platform (*manager*). Under these definitions, the model can be used to analyze online platforms that function as monitor-allocators. Examples of such platforms include Uber or Airbnb, who maintain relationships on both sides of the market, track provider performance, and direct consumers to high-performing providers. The role of online platforms as providers of relational trust is emphasized by legal scholars Henderson and Churi (2019) among others. Even though platforms often enter formal contracts with trading parties, there are also relational aspects in their interactions with traders that are not written into formal contracts or enforceable through courts. For instance,

[8]As of 2024, professional and business service firms accounted for 13% of U.S. employment.

[9]Ahn, Khandelwal and Wei (2011) estimate that intermediaries facilitated 20% of China's exports in 2005, while Bernard, Grazzi and Tomasi (2015) find that over 25% of Italian exporters act as intermediaries, accounting for 10% of national exports. See also Belavina and Girotra (2012) and Antràs and Chor (2022).

drivers may behave impolitely to riders in ways that are not observable to courts, and Uber has the discretion to exclude poorly behaving drivers from its platform. Our model shows how a platform may facilitate exchange by simultaneously coordinating allocations, tracking performance, and excluding drivers who behave poorly.[10]

4 Optimal Direct and Managed Relational Contracts

In this section we characterize optimal direct and managed relational contracts. We then analyze the buyer's choice between direct and managed relational contracts. This analysis explains why and when managed contracting may be preferred over direct contracting.

4.1 Direct Relational Contracts

We first focus on a single buyer who directly matches with a sequence of producers. The effect of other buyers is captured by the outside options available to the producers. We assume that the vacancies opened by other buyers are such that the pre-matching continuation value of a unmatched producer is stationary. Let $\overline{U} > 0$ denote the producer's pre-matching continuation value.

First, consider a buyer i in calendar time τ who is unmatched before producer matching. Suppose the buyer has demand and opens a vacancy with some continuation value U_{1i}. Since matching is frictionless, the vacancy is filled with certainty if and only if $U_{1i} > \overline{U}$.

Next, consider a buyer who is matched with a producer at the start of period τ and they are in period t of the current match. Further suppose that they do not separate after demand realization. The rest of period transpires as follows: first, the buyer pays its producer w, then

[10]Here we think of the function of the platform's algorithmic computer systems as similar to the allocative and monitoring function traditionally performed by human managers in professional service firms. To date, the idea of online platforms functioning as an incentive system has received little attention. Existing literature largely focuses on platform pricing in the presence of usage and membership externalities (Rochet and Tirole 2003, 2006) and the impact of platforms on consumer search (Brynjolfsson and Smith 2000; Ellison and Ellison 2009). A notable exception is Hagiu and Wright (2015), who model the organizational difference between vertical integration and multi-sided platforms as a difference in the allocation of decision rights, following Gibbons (2005).

the producer exerts effort e, and finally the producer buyer can terminate the relationship.

Following Board and Meyer-ter-Vehn (2015), we say that strategies are *contract-specific* if (1) the current wage w_t only depends on past wages and effort levels in the current match, $(w_1, e_1, ..., w_{t-1}, e_{t-1})$, (2) current effort e_t additionally depends on w_t, and (3) separation decisions additionally depend on e_t. Contract-specific strategies are thus independent of calendar time τ, the identity of the producer, or histories outside of the current match.

We define a *relational contract* as a subgame-perfect equilibrium in pure, contract-specific strategies where deviations from the equilibrium path are punished in the harshest possible way. In a direct match, this means that the producer shirks and quits if the buyer deviates from equilibrium pay $w_{i,t}$; similarly, the buyer fires the producer if he deviates from equilibrium effort $e_{i,t}$.

The post-matching continuation payoff for a matched producer, when $d_{i,t} = 1$, is:

$$U_{1i,t} = w_{1i,t} - c(e_{i,t}) + \delta\Big[(1-\alpha_{1i})U_{1i,t+1} + \alpha_{1i}(\beta_{i,t+1}U_{0i,t+1} + (1-\beta_{i,t+1})\overline{U})\Big], \tag{1}$$

and when $d_{i,t} = 0$, is:

$$U_{0i,t} = w_{0i,t} + \delta\Big[\alpha_{0i}U_{1i,t+1} + (1-\alpha_{0i})(\beta_{i,t+1}U_{0i,t+1} + (1-\beta_{i,t+1})\overline{U})\Big], \tag{2}$$

where $\beta_{i,t}$ denotes the buyer's probability of retaining the producer when $d_{i,t} = 0$.

Analogously, the post-matching continuation payoff for the buyer, when $d_{i,t} = 1$, is:

$$\Pi_{1i,t} = y_{i,t} - w_{1i,t} + \delta\Big[(1-\alpha_{1i})\Pi_{1i,t+1} + \alpha_{1i}(\beta_{i,t+1}\Pi_{0i,t+1} + (1-\beta_{i,t+1})\overline{\Pi}_{0i})\Big], \tag{3}$$

and when $d_{i,t} = 0$, is:

$$\Pi_{0i,t} = -w_{0i,t} + \delta\Big[\alpha_{0i}\Pi_{1i,t} + (1-\alpha_{0i})(\beta_i\Pi_{0i,t} + (1-\beta_i)\overline{\Pi}_{0i})\Big], \tag{4}$$

where $\overline{\Pi}_{0i}$ and $\overline{\Pi}_{1i}$ are the buyer's pre-matching continuation values when $d_{it} = 0$ and $d_{it} = 1$, respectively. Since there is an excess of producers in the frictionless matching market, the buyer can always successfully find a match, so $\overline{\Pi}_{1i} = \Pi_{1i,1}$ and $\overline{\Pi}_{0i} = \Pi_{0i,1}$.

The relational contract must satisfy the following incentive constraints for the producer:

$$U_{1i,t} \geq w_{1i,t} + \delta \overline{U}, \forall t, \tag{P-IC-e}$$

$$U_{1i,t} \geq \overline{U}, \forall t, \tag{P-IC1}$$

$$U_{0i,t} \geq \overline{U}, \forall t. \tag{P-IC0}$$

Constraint (P-IC-e) requires the producer to choose effort over shirking when the service is needed. Constraints (P-IC1) and (P-IC0) require that the producer remain with the current buyer when the demand is 1 or 0, respectively. If $\beta_{i,t} = 0$, equation (2) and constraint (P-IC0) do not apply, as the buyer immediately separates from the producer when demand is zero.

The relational contracts must also satisfy incentive constraints for the buyer:

$$\Pi_{1i,t} \geq \delta \Big[(1 - \alpha_{1i}) \overline{\Pi}_{1i} + \alpha_{1i} \overline{\Pi}_{0i} \Big], \tag{B-IC-w1}$$

$$\Pi_{0i,t} \geq \delta \Big[(1 - \alpha_{0i}) \overline{\Pi}_{0i} + \alpha_{0i} \overline{\Pi}_{1i} \Big], \tag{B-IC-w0}$$

$$\Pi_{1i,t} \geq \overline{\Pi}_{1i}, \tag{B-IC1}$$

$$\Pi_{0i,t} \geq \overline{\Pi}_{0i}. \tag{B-IC0}$$

Constraints (B-IC-w1) and (B-IC-w2) ensure that the buyer honors the payment to the producer.[11] Constraints (B-IC1) and (B-IC0) reflect the buyer's desire to retain the producer when there is a demand or not, respectively. As before, if $\beta_{i,t} = 0$, the term $\Pi_{0i,t}$ and constraint (B-IC0) do not apply, as the buyer would immediately separate from the producer.

The buyer's problem is to choose a relational contract that maximizes profit $\Pi_{1i,1}$, subject to (B-IC-w), (B-IC1), (P-IC-e), (P-IC1), as well as (B-IC0) and (P-IC0) whenever they apply.

[11] Since the producer exerts effort after the buyer makes the payment, the buyer would not receive the output should the buyer dishonor the payment to the producer. Therefore, there is no output or wage payment in period t on the right hand side of the constraint.

Stationarity

We now show that we can restrict to *stationary* relational contracts—where strategies are time-invariant within a match, with $e_{i,t} = e_i$, $w_{1i,t} = w_{1i}$, $w_{0i,t} = w_{0i}$, and $\beta_{i,t} = \beta_i$ for all t.

Lemma 1. *Assume stationary outside condition $\overline{U}$ and fix a non-stationary relational contract $(e_{i,t}, w_{1i,t}, w_{0i,t}, \beta_{i,t})_t$ with non-negative profit, $\Pi_{1i,1} \geq 0$. There is a stationary relational contract $(e_i^*, w_{1i}^*, w_{0i}^*, \beta_i^*)_t$ that generates a weakly higher profit for the buyer, $\Pi_{1i}^* \geq \Pi_{1i,1}$.*

Proof. All omitted proofs are in the appendix. □

Intuitively, buyers would like to back-load payments to incentivize effort. Indeed, the optimal contract would be back-loaded if the buyer's outside option were exogenous. However, the outside option here is endogenously determined. If new producers can be hired on back-loaded terms, then the buyer would wish to separate with expensive incumbent producers and replace them with new producers. This argument is originally due to Shapiro and Stiglitz (1984) and is formally shown by MacLeod and Malcomson (1989). Lemma 1 extends the argument to a setting with limited liability and demand fluctuations.

Lemma 1 does not fully rule out non-stationary buyer-optimal contracts. Given any stationary buyer-optimal contract, one can construct a non-stationary buyer-optimal contract by reducing pay and effort in the first period in a way that leaves profits constant. We focus on stationary firm-optimal contracts because, by Lemma 1, non-stationary contracts harm producers without benefiting buyers. Stationary contracts are also analytically tractable.

Two assumptions are made to reflect the large, anonymous nature of the matching market. First, we assume that buyers and producers cannot observe each other's behavior in past relationships. This is important: if producer 2 could observe a buyer's behavior with producer 1, then 2 could punish the buyer for unfairly firing 1, thereby enabling the buyer to profitably back-load 1's pay. Second, we consider equilibria in contract-specific strategies. If the firm and workers conditioned their behavior on calendar time or the identity of a worker, they could cut payments in the first period of the game for the first producers, without affecting incentives for subsequent producers. We rule out such strategies, since they require the first producer to believe they are special, which is inconsistent with our idea of a large anonymous matching market.

Another important assumption is that the payment w_t is nonnegative. This *limited liability* assumption is made for two reasons. First, it is realistic in many real-world settings. Second, it greatly simplifies the analysis. Because of limited liability, equilibrium pay is always zero during periods when demand is zero (see Lemma A.1). One can thus characterize each buyer's choice between stationary direct and managed contracts simply by comparing the buyer's per-period payment when demand is one.

In addition, we disallow bonus payments, which play an important role in MacLeod and Malcomson (1998) and Levin (2003). In Appendix A.2, we show that this assumption is not critical. The proof extends the argument for a similar result in Board and Meyer-ter-Vehn (2015): For any contract with bonuses, there is a weakly more profitable contract for buyers without bonuses. Intuitively, the buyer prefers to pay the producer at the latest possible moment before e_t. It is therefore weakly profitable to shift the bonus b_{t-1} into the next period's expected pay, such that an proportionately increased amount is paid only if the next period's demand is one and the producer is retained.

Optimal direct relational contract

Lemma 1 enables us to restrict our attention to buyer-optimal, contract-specific, stationary relational contracts. Let $C_i^D = (w_{1i}, w_{0i}, \beta_i)$ denote such a contract. In this contract, the producer's effort level is one if and only if $d_{it} = 1$, w_{1i} is the payment when $d_{it} = 1$, w_{0i} is the payment when $d_{it} = 0$, and $\beta_i \in [0, 1]$ is the probability that buyer i stays with the producer when the buyer's demand is zero. The time subscript is now dropped.

The buyer-optimal direct contract is obtained by choosing w_{1i}, w_{0i}, and β_i to maximize Π_{1i}, subject to (B-IC-w), (B-IC1),(P-IC-e), (P-IC1), as well as (B-IC0) and (P-IC0) if $\beta_i > 0$.

Lemma 2. *Suppose an optimal direct relational contract exists. Under this relational* *contract, if*

$$\frac{(1-\delta)\overline{U}}{c} > \frac{\alpha_{0i}}{1-\alpha_{1i}}, \tag{5}$$

then the buyer pays

$$w_{DS}(\alpha_i) = \left(\frac{1}{\delta}\frac{1}{1-\alpha_{1i}}\right)c + (1-\delta)\overline{U}. \tag{6}$$

when demand is one and separates from the producer when demand switches to zero. Otherwise, the buyer pays

$$w_{DI}(\alpha_i) = \left(\frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_{0i}}{(1-\alpha_{1i})+\frac{\delta}{1-\delta}\alpha_{0i}}\right)c + \left(\frac{1+\frac{\delta}{1-\delta}\alpha_{0i}}{(1-\alpha_{1i})+\frac{\delta}{1-\delta}\alpha_{0i}} - \delta\right)\overline{U} \tag{7}$$

when demand is one, remains matched with the idle producer when demands switches to zero, and pays zero until demand switches back to one.

Lemma 2 shows that the optimal direct contract features a buyer who either always separates from a producer or always retains him when their demand switches to zero. According to Equation (5), direct contracts with separation dominates direct contracts with idleness when (a) the producer's continuation value when unmatched $\overline{U}$ is high, (b) the buyer has a smaller α_{0i}, so a longer period of idleness is expected, and (c) the buyer has a larger α_{1i}, so a shorter period with service demand is expected.

The payment to producers in a direct contract, $w_D(\alpha_i) \equiv \min\{w_{DS}(\alpha_i), w_{DI}(\alpha_i)\}$, is increasing in α_{1i}. This is because when business needs are shorter-lived, the producer faces a higher chance of either separating or becoming idle, so the future surplus in the relationship is smaller. A higher incentive rent is therefore needed to incentivize producer effort.

4.2 Managed Relational Contracts

In managed matches, buyers enter relational contracts with managers, who are delegated the responsibility of motivating producers. The manager fulfills the buyer's demand by entering relational (sub-)contracts with a large number of producers and assigning producers to buyers according to demand realization. As we shall show, by aggregating fluctuating buyer demand and coordinating buyer-producer assignments, managers can reduce the cost of motivating producer performance.

To demonstrate this, we first analyze the (sub-)contracts that managers enter with producers. We argue here that unlike buyers, who have fluctuating demand, managers have constant demand for services. Why? In equilibrium, each manager randomly matches with a positive measure of buyers drawn from the same type distribution. Therefore, by the law of large numbers, the total measure of demand that managers receive from buyers is unchanging over time. Given this stable demand for services, the measure of producers that each manager contracts with is equal to the expected measure of demand realizations. Each matched producer in these contracts exert effort in every period.

For tractability, we first assume that each manager enters manager-optimal, contract-specific, and stationary contracts with producers to meet buyer demand. This assumption can be justified by the same argument as in the previous subsection.

Second, we assume that producers are matched in a large anonymous market. This assumption implies that managers will, with positive probability, immediately rematch with producers from whom they just separated, but will not know their histories. This assumption is admittedly unnatural, but it significantly increases analytic tractability, since it enables us to reuse the machinery that we previously developed for direct relational contracts to analyze subcontracts between managers and producers. It is a reasonable approximation when the number of managers is large, but it is problematic when the number of managers is small.

This second assumption can be viewed as yielding a conservative upper bound on the incentive pay required in managed contracts. If matching is not anonymous, managers will never rematch with a producer that it previously fired. Managers can then threaten punishments with greater severity to underperforming producers, since each manager contracts with a nonzero measure of producers. As such, the incentive pay required would be lowered. This is the mechanism highlighted by Biglaiser and Friedman (1994), where intermediaries offer lower incentive rents to sellers purely due to their market share.[12]

[12] Biglaiser and Friedman (1994) show middlemen with reputations have a larger incentive to monitor and are in a better position to learn about quality than an ordinary buyer does because they buy a larger proportion

Our assumption of anonymity shuts down this additional *market-share* benefit of managed contracts, allowing us to instead focus on the manager's *demand-smoothing* role in fluctuating environments.

Given these assumptions, the payment for the producer is $w_{1i} = w_M$ in every period. By the same logic as Lemma 2, we have that

$$w_M = \frac{c}{\delta} + (1-\delta)\overline{U}. \tag{8}$$

The demand-smoothing benefit of managed contracts is apparent from Equation (8). Since the managers can aggregate fluctuating demand across their buyers, their payments to producers (w_M) are lower than buyer payments to producers under direct contracts (w_D). Mathematically, $w_D(\alpha_i) \geq w_M$ for all α_i, where equality holds if and only if $\alpha_{1i} = 0$.

Next, we consider how producers are assigned to buyers in each period under the managed contract. Note that buyers are indifferent between any assignment of producers where the assigned producer exerts effort, since producers are identical in our model. Producers are also indifferent between any assignment of buyers where the managers offer the same level of payment. Since managers require producers to exert effort and provide the same compensating payment in every period, buyers and producers have the same payoffs in any assignment where producers are matched with buyers with positive demand in every period. There are an infinite number of such assignments, and any such assignment is optimal. The multiplicity of optimal assignments gives rise to a need for each manager to coordinate assignments between their matched buyers and producers. For simplicity, we will not explicitly model the coordination process and instead assume that an optimal assignment is chosen.

Now, we characterize the contract between buyers and managers. The post-matching

of the producers' goods (see also Bardhan, Mookherjee and Tsumagari 2013). Unlike their model, managers do not have reputations in our baseline model; they simply enter relational contracts on both sides of the market.

continuation payoffs for the manager, when the service is and is not needed, respectively, are

$$V_{1i,t} = p_{1i,t} - w_M + \delta\Big[(1-\alpha_{1i})V_{1i} + \alpha_{1i}(\beta_{i,t}V_{0i,t} + (1-\beta_{i,t})\overline{V})\Big], \tag{9}$$

and

$$V_{0i,t} = p_{0i,t} + \delta\Big[\alpha_{0i}V_{1i,t} + (1-\alpha_{0i})(\beta_{i,t}V_{0i,t} + (1-\beta_{i,t})\overline{V})\Big], \tag{10}$$

where $\overline{V}$ is a manager's continuation value after separating from a buyer. The relevant incentive constraints for the manager, similar with those for a producer, are

$$V_{1i,t} \geq p_{1i,t} + \delta\overline{V}, \tag{M-IC-w}$$

$$V_{1i,t} \geq \overline{V}, \tag{M-IC1}$$

$$V_{0i,t} \geq \overline{V}. \tag{M-IC0}$$

For the buyer, the continuation payoffs and incentive constraints are the same as in the direct contract, except that the payments w_{0i} and w_{1i} to the producer are replaced with service fee payments p_{0i} and p_{1i} to the manager. For concision, we omit these conditions, which simply repeat (B-IC-w), (B-IC-1), and (B-IC0).

Unlike producer matches, we assume that matching between buyers and managers is *not* anonymous. If a manager separates from a buyer, it cannot match with a new buyer, because all other potential buyers are matched with some manager and will not become unmatched on the equilibrium path. The manager also cannot re-match with the initial buyer, since manager-buyer matching is assumed to not be anonymous. Therefore, $\overline{V} = 0$ on the equilibrium path.

The assumption of non-anonymity is important for our result that managed contracts are sometimes cheaper than direct contracts. Suppose instead that manager matching is anonymous. In this case, a buyer who separates from a manager will immediately rematch with the same manager with positive probability, but the buyer will not know the identity or history of this manager. A larger incentive rent will then be needed to motivate the manager.

Once again, we assume that buyers enter manager-optimal, contract-specific, and stationary contracts with manager. Let $C_i^M = (p_{1i}, p_{0i}, \beta_i)$ denote such a managed relational contract. Here p_{1i} and p_{0i} are the service fees when the buyer needs and does not need the service, respectively. If the buyer deviates from the specified service fee, the manager does not assign a producer to the buyer and separates from the buyer. If instead the producer assigned by the manager deviates from the specified effort, the buyer separates from the manager after production. The optimal managed contract maximizes Π_{1i} subject to these incentive compatibility constraints.

Lemma 3. *Suppose an optimal managed contract exists. Under this contract, the buyer always retains the manager, pays zero service fees to the manager when there is no demand, and when there is demand, she pays the manager a service fee equal to*

$$p(\alpha_i) = \left(\frac{1}{\delta} \frac{1 + \frac{\delta}{1-\delta}\alpha_{0i}}{(1 - \alpha_{1i}) + \frac{\delta}{1-\delta}\alpha_{0i}} \right) w_M. \tag{11}$$

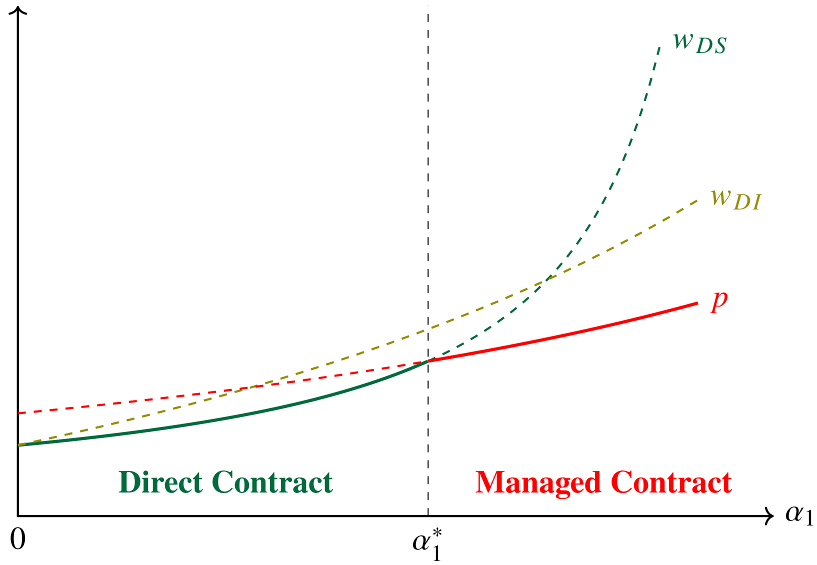
Lemma 3 shows that the cost of managed contract is a form of double marginalization. Note that manager pay $p(\alpha_i)$ is the product of $\lambda(\alpha_i) = \frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_{0i}}{(1-\alpha_{1i})+\frac{\delta}{1-\delta}\alpha_{0i}}$ and producer pay w_M. Here $\lambda(\alpha_i)$ can be thought of as a manager markup. Furthermore, as shown in Equation (8), w_M is elevated above the producer's cost of effort. In other words, both the manager and producer are paid rents so that both are incentivized to honor their contractual obligations.[13]

4.3 Optimal Contractual Choice

Having characterized direct and managed contracts, we now characterize when buyers choose managed contracts. We focus on buyers who have the same α_{0i}, which is the rate at which a

[13]This form of double marginalization is different the traditional textbook type. In the classic literature in industrial organization, double marginalization occurs when both upstream and downstream parties possess market power and use restrictive linear contracts. It can thus be eliminated by vertical integration of pricing decision rights, resale-price maintenance, or non-linear pricing (Tirole 1988). In our model, double marginalization instead arises from the non-verifiability of contract performance and the non-transferability of production and monitoring capabilities, as in models of middleman margins (e.g., Biglaiser and Friedman 1994; Bardhan, Mookherjee and Tsumagari 2013). Conventional remedies do not eliminate this form of double marginalization.

Figure 4: Service cost under direct and managed relational contracts



Note: α_1 is the probability that the buyer's demand switches from 1 to 0.

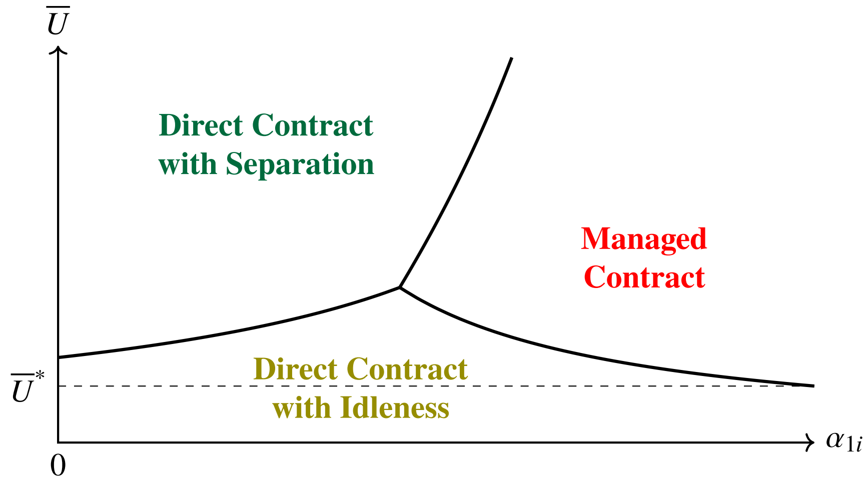
buyer's demand changes from zero to one. We examine how the optimal contractual choice depends on α_{1i}, the rate at which the buyer's demand changes from one to zero, and $\overline{U}$, the producer's outside option, which in equilibrium reflects whether the producer market is tight and whether it is easy for producers to rematch.

To compare the two, it suffices to compare the payment $w_D(\alpha_i)$ under direct contracting and the service fee $p(\alpha_i)$ under managed contracting. This is because the buyer pays nothing when their demand is zero and can always match with a producer when her demand is one.

Figure 4 provides a graphical illustration. For a buyer with stable demand, i.e., for whom $\alpha_{1i} = 0$, managed contracting is strictly more expensive than direct contracting because of double marginalization. To obtain high-quality services, the buyer needs to pay additional rent to the manager. However, there is no benefit to managed contracts, since demand is stable, so the producer is paid the same incentive rent under direct contracting.

The advantage of managed contracting over direct contracting becomes larger when business needs are short-term. As α_{1i} becomes larger, producers must be paid elevated payments in order for them to exert effort. Since managers can reassign producers across

Figure 5: Optimal contractual choice given model parameters



buyers, producers never separate or become idle, so the cost of incentivizing producers is lowered. To see this mathematically, note that $w_{DS}(\alpha_i)$ approaches infinity as α_{1i} approaches one. Furthermore, $p(\alpha_i)$ increases in α_{1i} less steeply than $w_{DI}(\alpha_i)$ if c is relatively small and $\overline{U}$ is relatively large. Therefore, when rematching is easy, the manager operates a more cost-efficient internal producer market by having long-standing relational contracts on both sides of the market.

Figure 5 graphically shows the optimal relational contracts as a function of α_{1i} and $\overline{U}$. In this figure, provided $\overline{U} > \overline{U}^*$, then there exists a cutoff value such that managed contracts dominate if and only if α_{1i} is sufficiently large. If $\overline{U} < \overline{U}^*$, then the producer's pre-matching continuation value is low, so it is optimal to retain them and keep them idle until demand returns.

The following proposition formalizes this result.

Proposition 1. *Take any set of buyers i with the same α_{0i} for whom an optimal contract exists. There exists $\overline{U}^* > 0$ such that:*

1. *If $\overline{U} > \overline{U}^*$, there exists $\alpha_1^*(\alpha_{0i}, \overline{U}) \in (0, 1)$ such that a managed contract is optimal if and only if $\alpha_{1i} \geq \alpha_1^*(\alpha_{0i}, \overline{U})$;*

2. *If $\overline{U} < \overline{U}^*$, a direct contract is optimal for all i in this set.*

5 Steady-State Equilibrium

In this section, we show that there exists a unique steady-state equilibrium. We then explore the model's empirical implications. First, we compare the outcomes of managed producers and directly contracted producers in equilibrium. Second, we compare the outcomes and welfare of buyers and producers in economies with and without managers.

5.1 Deriving the Equilibrium

We say that the economy is in a steady state when (1) the number of producers in the economy and the distribution of contracts in the matching market are unchanging across periods and (2) each buyer's demand evolves according to its stationary distribution. By the properties of Markov processes, the steady-state probability that a buyer i has positive demand in any period is equal to $\pi_i = \alpha_{0i}/(\alpha_{0i} + \alpha_{1i})$.

At the start of each period, some buyers and managers are unmatched and directly offer direct contracts to unmatched producers. Let $\mathcal{I}_{DS}$ denote the set of buyers who enter direct contracts that end when their demand switches to zero. For a buyer $i \in \mathcal{I}_{DS}$, the steady-state probability that they offer new contracts is

$$v_i = (1 - \pi_i)\alpha_{0i}. \tag{12}$$

Let $\mathcal{I}_{DI}$ be the set of buyers who directly contract with producers in contracts that continue when demand switches. Let $\mathcal{I}_M$ be the set of buyers who contract with managers. For both types of contracts, producers never separate. Therefore, for any buyer $i \in \mathcal{I}_{DI} \cup \mathcal{I}_M$, the steady-state probability that they offer new contracts is $v_i = 0$. The total measure of new contracts offered by buyers and managers in the producer market is given by

$$v = \int_{\mathcal{I}_{DS}} v_i dF. \tag{13}$$

The steady-state measure of producers in direct contracts, n_B, is given by

$$n_B = \int_{\mathcal{I}_{DS}} \pi_i dF + \int_{\mathcal{I}_{DI}} dF. \tag{14}$$

The steady-state measure of producers in managed contracts, n_M, is given by[14]

$$n_M = \int_{\mathcal{I}_M} \pi_i dF. \tag{15}$$

The measure of unmatched producers after matching is

$$n_N = n - n_B - n_M. \tag{16}$$

If y and C are sufficiently large and the buyer type distribution F has full support on $[0, 1] \times [0, 1]$, the value of entering either direct or managed contracts is higher than the value of being unmatched. This implies that there is an excess of producers who enter the producer market, so $n_N > 0$.

Since matching is frictionless, all contracts offered in the producer market are immediately filled. The total measure of unmatched producers before matching, u, is given by

$$u = v + n_N. \tag{17}$$

Matching is random and contract offers are never made to matched producers, so the Bellman equation for an unmatched producer is given by

$$\overline{U} = \int_{\mathcal{I}_{DS}} \frac{v_i}{u} U_{DS}(\alpha_i) dF + \left(1 - \frac{v}{u}\right) \overline{U}, \tag{18}$$

where $U_{DS}(\alpha_i) = \frac{1}{1-\delta(1-\alpha_{1i})}(w_{DS}(\alpha_i) - c + \delta\alpha_{1i}\overline{U})$. Note that the continuation value of being unmatched ($\overline{U}$) consists of two components. The first term reflects the rent from matching with an unmatched buyer. The second term reflects the value of remaining unmatched.

[14]Here π_i enters the integral since the manager aggregates the fluctuating demand of buyers.

To complete the model, we solve for the steady-state number of producers that enter the economy. By assumption, producers enter the economy at the beginning of each period by paying an entry cost C. Entry drives down the likelihood that producers are matched, so they enter only until the continuation value of being unmatched in the labor market equals their entry cost. This yields the following condition:

$$\overline{U} = C. \tag{19}$$

We can now define an equilibrium in our economy.

Definition 1. *A steady-state equilibrium is a distribution of relational contracts offered by each buyer and manager such that:*

1. *All relational contracts are offerer-optimal, contract-specific, and stationary;*
2. *Each player's pre-matching continuation value is determined by steady-state transition probabilities and frictionless and random matching via Equation* (18)*;*
3. *The measure of producers in the economy is derived from the producer entry condition, given by Equation* (19)*.*

Proposition 2. *There exists a unique steady-state equilibrium. If y and C are sufficiently high and F has full support on* $[0, 1] \times [0, 1]$*, then a non-zero measure of buyers enter direct contracts, while another non-zero measure of buyers enter managed contracts.*

Proof. By Equation (19), $\overline{U}$ equals the entry cost C. Given $\overline{U}$, we can compare the values of $w_{DS}(\alpha_i)$, $w_{DI}(\alpha_i)$, $p(\alpha_i)$, and y for each i using Equations (6), (7), and (11) to determine $\mathcal{I}_{DS}$, $\mathcal{I}_{DI}$, and $\mathcal{I}_M$. Having derived these, we can obtain unique values for v_i, v, n_B, and n_M from Equations (12), (13), (14), and (15). We plug these into Equation (18) to solve for a unique value for u. Plugging u into Equations (16) and (17) then yields a unique value for n. The desired statement then follows from Proposition 1. □

5.2 Empirical Implications

Having shown that there exists a unique steady-state equilibrium in this exchange economy, we can now explore the model's predictions regarding equilibrium behavior.

Effects of Managerial Coordination on Producers

How does managerial coordination affect the pay, separation rates, and idleness of producers? We focus on the interesting case where all three contract types arise in equilibrium.[15] We define a producer's *separation rate* as the probability that a matched producer becomes unmatched at the start of the next period. We define *idleness* as the steady-state probability that a producer is matched but does not exert effort. We derive four findings.

Corollary 1. *Directly contracted producers have (1) higher average pay, (2) more dispersed pay, (3) higher separation rates, and (4) higher idleness than managed producers.*

The first result follows from the fact that $w_D(\alpha_i) > w_M$ for all buyers i with fluctuating demand. The second follows from the fact that $w_D(\alpha_i)$ takes on different values depending on α_i, which is heterogeneous across buyers, while w_M is constant. The final two follow from the fact that directly contracted producers may separate from buyers or become idle when demand changes, while managed producers are always reallocated among the manager's clients, so they never separate nor become idle.

The predictions broadly align with recent evidence. Workers managed by professional service firms earn lower and more compressed wages (Dube and Kaplan 2010; Goldschmidt and Schmieder 2017; Drenik et al. 2023). Some evidence also suggests that they have lower hazard into unemployment than comparable direct employees (Guo, Li and Wong 2024).

Trade and Welfare with and without Managerial Coordination

How does the presence of managers alter market equilibrium and the welfare of buyers and producers? Felix and Wong (2024) show that Brazil's 1993 legalization of professionally

[15]Specifically, we assume that y, C and F are such that $|\mathcal{I}_M|, |\mathcal{I}_{DI}|, |\mathcal{I}_{DS}| > 0$. If F has full support on $[0, 1] \times [0, 1]$, then by Proposition 1 there exists y and C such that this holds.

managed security firms persistently increased security guard employment, but displaced incumbent security guards from high-wage direct employers.

To explore the same question theoretically, we consider two economies: one with managers and one without. The introduction of managers causes three types of buyers to switch to managed contracts: buyers who do not initially consume services, buyers who initially choose direct contracts with idleness, and buyers who initially choose direct contracts with separation. We denote the three subsets of switching buyers as $\mathcal{S}_0$, $\mathcal{S}_I$, and $\mathcal{S}_S$. We assume that there is a positive measure of switchers who initially do not consume, and that there is a positive measure of switchers who directly contract initially.[16]

In both economies, producers are assumed to freely enter at cost C. As such, the introduction of managers does not alter the payments received by producers who remain directly matched to same buyers. It only reduces the payments received by producers who are become matched to managers instead of buyers. It follows that:

Corollary 2. *When managers are introduced into an economy, the payoffs of all buyers weakly increase and the measure of buyers who receive services increases. The mean pay per unit effort received by producers falls. The total measure of matched producers increases if and only if* $\int_{\mathcal{S}_0} \pi_i dF > \int_{\mathcal{S}_I} (1 - \pi_i) dF$.

These predictions are intuitive. The introduction of managers benefits buyers with fluctuating demand, but it hurts incumbent producers who were initially earning pay premiums. As such, it unambiguously increases the measure of buyers who receive services. However, it is ambiguous whether the measure of matched producers ($n_M + n_B$) increases or falls. On one hand, managers enable more buyers to afford services, which increases demand for producers. On the other, managers reduce idleness, so fewer producers are needed. The relative magnitude of these two effects depends on the distribution of buyer types.

[16]In other words, y, C and F are such that $|\mathcal{S}_0| > 0$ and $|\mathcal{S}_I \cup \mathcal{S}_S| > 0$. Assuming indifferent buyers do not switch, this requires that there exists $\alpha_i, \alpha_i' \in \operatorname{supp} F$ such that $y > w_D(\alpha_i) > p(\alpha_i)$ and $w_D(\alpha_i') > y > p(\alpha_i')$. By Proposition 1, if F has full support on $[0, 1] \times [0, 1]$, then there exist y and C such that this holds.

6 Extensions

This section extends the model to study the relationship between managerial coordination and producer specialization. We show that buyers sort into managerial coordination when both demand is volatile and needs are specialization. We also characterize how managerial coordination impacts the division of labor. Further extensions incorporating manager reputation concerns and an exogenous producer population are presented in the Appendix and briefly discussed below.

6.1 Producer Specialization

We consider a multi-task economy with a measure of buyers who have unit demand in every period, for either one of two tasks $d_{it} \in \{A, B\}$. Each buyer's demand switches from one task to another task with some symmetric probability α_i at the start of each period.[17] There is also an excess measure of producers indexed by j who choose whether to become either specialists in one of two activities needed by buyers or a generalist with middling skill in both services. Let $\phi_j \in \{A, B, G\}$ denote the chosen type of the producer, where A and B refer to specialists and G refers to the generalist. The output depends on the buyer's demand d_{it}, the producer's type ϕ_j, and the producer's chosen effort e_{jt}, and is given by

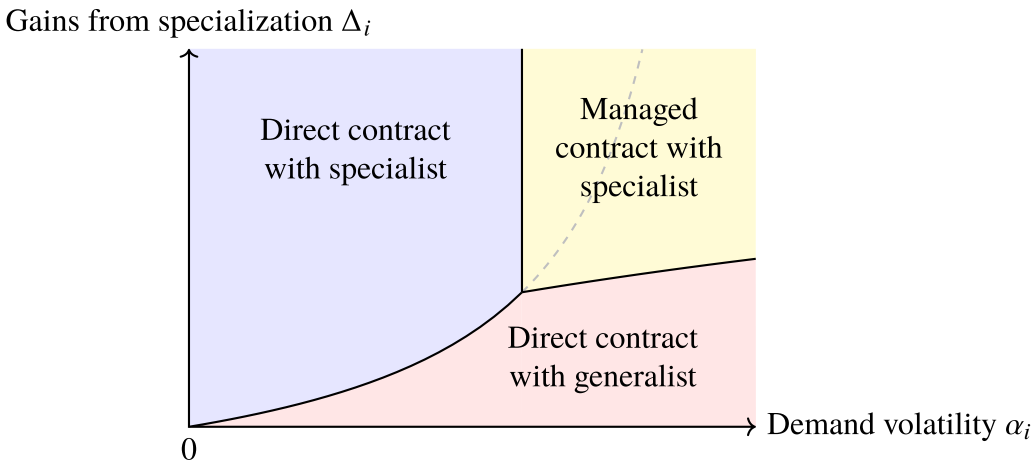
$$y_{ijt} = \left[y \cdot \mathbf{1}\{\phi_j = G\} + (y + \Delta_i) \cdot \mathbf{1}\{\phi_j = d_{it}\}\right] e_{jt},$$

where $\Delta_i > 0$ denotes the buyer-specific *gains from specialization*. If the specialist exerts effort, output is high when demand and the producer's type are well-matched, but low when they are not. Output is always middling for generalists who exert effort.

As before, neither pay w nor effort e are contractible and must be incentivized through relational contracts. We assume that output y is sufficiently large so that buyers are always able to receive positive profit by directly contracting with a generalist, so there are never buyers who do not receive services. We also assume that each producer's entry cost C is

[17] This is essentially simplifying the model in Section 3 by assuming $\alpha_{Ai} = \alpha_{Bi} = \alpha_i$.

Figure 6: Optimal contractual choice in the presence of gains from specialization



sufficiently large so that specialist producers never remain in a contract but become idle when the demand of their buyer changes. These assumptions allow us to focus on each buyer's choice between direct contracting with a generalist, direct contracting with a specialist who is never idle, and managed contracting. There are K managers for each task. Each manager can only monitor producers specializing in that task and are randomly matched with buyers who offer managed contracts.[18]

Drivers of Managerial Coordination in a Multi-task Economy. Abraham and Taylor (1996) find that establishments with cyclical demand and specialized needs are more likely to outsource accounting services to professionally managed firms. Similar patterns are found in our model, as visualized in Figure 6 and stated formally in the following proposition.

Proposition 3. *A unique steady-state equilibrium exists in a multi-task economy. Buyer i chooses managed contracts if and only if their demand volatility α_i and gains from specialization Δ_i are both sufficiently large.*

Division of labor with and without managers. We next compare multi-task economies with and without managers. We assume that if managers are present, a non-zero measure of

[18] This setup implicitly assumes that each manager specializes in monitoring one type of task.

buyers choose managed contracts and the conditional distribution of α_i given Δ_i has positive support on $[0, 1]$.

Corollary 3. *When managers are introduced into a multi-task economy, the measure of specialist producers increases and the measure of unmatched producers falls.*

The intuition is as follows. The dashed curve in Figure 6 shows the boundary between direct contracting with a specialist and with a generalist when managers are absent. When managers are present, part of the demand for directly contracted generalists is replaced with demand for managed specialists. Overall demand for specialists rises. In response, more producers choose to become specialists. Correspondingly, there are fewer direct contracts with elevated pay, so fewer producers enter, and the measure of unmatched producers falls.

6.2 Manager Reputational Concerns

Appendix B extends the model to allow managers to be disciplined through reputation as well as termination. In a multi-task economy, low effort by a producer under a given manager is publicly observed by other buyers with probability γ, with γ capturing the ease of word-of-mouth communication (e.g., via the Internet or social media). A higher γ tightens the manager's incentive constraint, reduces the markup required to keep the manager honest, and thus lowers the unit cost of managed services. Consequently, as γ increases, more buyers choose managed contracts, the measure of managed and specialist producers rises, and the measure of unmatched producers falls in the unique steady-state equilibrium.

6.3 Exogenous Producer Population

Appendix C considers a version of the baseline model where the number of producers $n > 1$ is fixed rather than determined by entry. With a fixed producer pool, managers affect the producer outside option $\overline{U}$ by changing aggregate demand for producers, so equilibrium wages and contract choices adjust endogenously. The appendix shows there is a unique steady-state equilibrium and that, if n is large enough, the market becomes very slack,

$\overline{U}$ approaches zero, buyers optimally use direct contracts with idleness, and managerial coordination disappears. Moreover, when managers are introduced with fixed n, they have both direct effects (buyers switching to managed contracts and lowering producer rents) and indirect effects through the induced change in $\overline{U}$, which can raise or lower all producers' pay depending on how many previously unmatched buyers become matched.

7 Conclusion

Economists have long puzzled over why the *visible hand* of managers often replace the *invisible hand* of markets in allocating resources. This paper sheds new light on the puzzle by developing a repeated-game model of an exchange economy in which managers coordinate allocations in decentralized matching markets. We show that managerial coordination reduces the cost of incentivizing performance in fluctuating environments. However, managerial coordination involves a trade-off between demand aggregation and double marginalization. Consequently, managers only partially replace anonymous matching markets in determining assignments between buyers and producers. In equilibrium, buyers sort into managerial coordination and decentralized exchange depending on their demand volatility and their gains from specialization. Managerial coordination alters aggregate output, the division of labor, matching market flows, and the distribution of economic rents.

A key contribution of our work is to show that managerial coordination is efficient in exchange economies, even in the absence of many factors previously hypothesized to be important, including contract-writing costs, bargaining frictions, search frictions, and adverse selection. The predictions of the model are not only applicable to a variety of organizational forms, they are also realistic when compared to recent empirical findings on professional service firms. However, the model here is rudimentary in many ways. Further analysis of endogenous relational contracting networks in decentralized matching markets, as attempted in this paper, may be a promising avenue for shedding new light on the microeconomic logic underlying the equilibrium behavior of economic organizations.

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Appendix

A Model with Endogenous Producer Entry

A.1 Proof of Stationarity, Lemma 1

We omit the subscript i in the proofs. To show that it is without loss of generality to focus on stationary relational contracts, we now construct a stationary relational contract $(e^*, w_1^*, w_0^*, \beta^*)$ based on any potentially non-stationary relational contract $(e_t, w_{1,t}, w_{0,t}, \beta_t)_t$ that generates a weakly higher continuation payoff for the buyer when starting a relationship, i.e., $\overline{\Pi}_1^* \geq \overline{\Pi}_1$.

Under $(e_t, w_{1,t}, w_{0,t}, \beta_t)_t$, the producer's and the buyer's continuation payoffs in a match are

$$U_{1,t} = w_{1,t} - c(e_t) + \delta\Big[(1-\alpha_1)U_{1,t+1} + \alpha_1(\beta_t U_{0,t+1} + (1-\beta_t)\overline{U})\Big],$$

$$U_{0,t} = w_{0,t} + \delta\Big[\alpha_0 U_{1,t+1} + (1-\alpha_0)(\beta_t U_{0,t+1} + (1-\beta_t)\overline{U})\Big],$$

$$\Pi_{1,t} = y(e_t) - w_{1,t} + \delta\Big[(1-\alpha_1)\Pi_{1,t+1} + \alpha_1(\beta_t \Pi_{0,t+1} + (1-\beta_t)\overline{\Pi}_0)\Big],$$

$$\Pi_{0,t} = -w_{0,t} + \delta\Big[\alpha_0 \Pi_{1,t+1} + (1-\alpha_0)(\beta_t \Pi_{0,t+1} + (1-\beta_t)\overline{\Pi}_0)\Big].$$

The incentive constraints for the producer and the buyer are:

$$U_{1,t} \geq w_{1,t} + \delta\overline{U},$$

$$U_{1,t} \geq \overline{U},$$

$$U_{0,t} \geq \overline{U},$$

$$\Pi_{1,t} \geq \delta(\alpha_1\overline{\Pi}_0 + (1-\alpha_1)\overline{\Pi}_1),$$

$$\Pi_{1,t} \geq \overline{\Pi}_1,$$

$$\Pi_{0,t} \geq \overline{\Pi}_0.$$

Now we construct $(e^*, w_1^*, w_0^*, \beta^*)$. First, suppose $e_{t'} = 1$ in some period t' and let $e^* = e_{t'} = 1$ (t' exists since otherwise the original relational contract will be a stationary in the effort with effort being 0 in all periods). Second, let $U_1^* = U_{1,t'}$ and $U_0^* = U_{0,t'}$. Finally, let $\beta^* = \beta_{t'}$ and set the payments w_1^* and w_0^* such that they satisfy:

$$U_1^* = w_1^* - c + \delta\Big[(1-\alpha_1)U_1^* + \alpha_1(\beta^* U_0^* + (1-\beta^*)\overline{U})\Big],$$

$$U_0^* = w_0^* + \delta\Big[\alpha_0 U_1^* + (1-\alpha_0)(\beta^* U_0^* + (1-\beta^*)\overline{U})\Big].$$

The buyer's continuation payoffs are thus

$$\Pi_1^* = y - w_1^* + \delta\Big[(1-\alpha_1)\Pi_1^* + \alpha_1(\beta^*\Pi_0^* + (1-\beta^*)\overline{\Pi}_0)\Big],$$

$$\Pi_0^* = -w_0^* + \delta\Big[\alpha_0\Pi_1^* + (1-\alpha_0)(\beta^*\Pi_0^* + (1-\beta^*)\overline{\Pi}_0)\Big].$$

Under this new stationary relational contract, the producer's incentive constraints are satisfied. Indeed, the first incentive constraint requires

$$\delta\Big[(1-\alpha_1)(U_1^* - \overline{U}) + \alpha_1\beta^*(U_0^* - \overline{U})\Big] \geq c,$$

as $\delta\Big[(1-\alpha_1)(U_{1,t'} - \overline{U}) + \alpha_1\beta_{t'}(U_{0,t'} - \overline{U})\Big] \geq c(e_{t'})$ at t'. Meanwhile, we know

$$U_1^* = U_{1,t'} \geq \overline{U},$$

$$U_0^* = U_{0,t'} \geq \overline{U},$$

The buyer's incentive constraints are satisfied since the relational contract is stationary, which implies $\Pi_1^* = \overline{\Pi}_1$ and $\Pi_0^* = \overline{\Pi}_0$ in this frictionless market with excess supply of producers.

To show $\overline{\Pi}_1^* \geq \overline{\Pi}_1$, given that $\overline{\Pi}_1^* = \Pi_1^*$ and $\Pi_{1,t'} \geq \overline{\Pi}_1$, we need only to show $\Pi_1^* \geq \Pi_{1,t'}$. We do so by comparing the joint surpluses $\Pi_1^* + U_1^*$ and $\Pi_{1,t'} + U_{1,t'}$. It is obvious that

$\Pi_1^* + U_1^* \geq \Pi_{1,t'} + U_{1,t'}$, since under the stationary relational contract, effort is motivated and thus the flow payoffs are maximized in each period when there is demand. Given that $U_1^* = U_{1,t'}$ by construction, we have $\Pi_1^* \geq \Pi_{1,t'}$.

A.2 Irrelevance of bonus payments

Based on the relational contract with bonus payments $(e_t, w_{1,t}, b_{1,t}, w_{0,t}, b_{0,t}, \beta_t)_t$, we can construct a new relational contract without bonus payments that generates the same continuation payoffs for the buyer, $(\hat{w}_{1,t}, \hat{w}_{0,t}, \hat{\beta}_t)_t$, by shifting the bonus payments in the last period to the next period with adjusting to discounting and potential separation.

Specifically, let $\hat{w}_{1,t} = w_{1,t} + \frac{(1-\alpha_0)\frac{b_1}{\delta} - \alpha_1 \frac{b_0}{\delta}}{1-\alpha_0-\alpha_1}$, $\hat{w}_{0,t} = w_{0,t} + \frac{(1-\alpha_1)\frac{b_0}{\delta} - \alpha_0 \frac{b_1}{\delta}}{1-\alpha_0-\alpha_1}$, and $\hat{\beta}_t = \beta_t$, $\forall t$. By doing so, all the incentive constraints are satisfied since the following conditions are satisfied:

$$-b_{1,t}+\delta\Big[(1-\alpha_1)\Pi_{1,t+1}+\alpha_1(\beta_t\Pi_{0,t+1}+(1-\beta_t)\overline{\Pi}_0)\Big] = \delta\Big[(1-\alpha_1)\hat{\Pi}_{1,t+1}+\alpha_1(\hat{\beta}_t\hat{\Pi}_{0,t+1}+(1-\hat{\beta}_t)\overline{\Pi}_0)\Big]$$

$$-b_{0,t}+\delta\Big[\alpha_0\Pi_{1,t+1}+(1-\alpha_0)(\beta_t\Pi_{0,t+1}+(1-\beta_t)\overline{\Pi}_0)\Big] = \delta\Big[\alpha_0\hat{\Pi}_{1,t+1}+(1-\alpha_0)(\hat{\beta}_t\hat{\Pi}_{0,t+1}+(1-\hat{\beta}_t)\overline{\Pi}_0)\Big].$$

Under this relational contract without bonus payments, the buyer's continuation payoffs stay the same as the original relational contract with bonus payments. It is therefore without loss to focus on relational contracts without bonus payments.

A.3 Implications of Limited Liability

We establish the following two lemmas.

Lemma A.1. *Suppose an optimal bilateral relational contract exists. Under this contract, if $\beta > 0$, then the following three conditions hold: 1) $w_0 = 0$, 2) both (B-IC1) and (B-IC0) bind, and 3) (P-IC1) is slack.*

Proof. Prove by contradiction. Suppose that $w_0 > 0$. Since there is an excess of producers in the matching market, $\overline{\Pi}_1 = \Pi_1$. So $\Pi_0 = -w_0 + \delta\left[\alpha_0\Pi_1 + (1-\alpha_0)(\beta\Pi_0 + (1-\beta)\overline{\Pi}_0)\right] =$

$-w_0+\delta\left[\alpha_0\overline{\Pi}_1+(1-\alpha_0)(\beta\Pi_0+(1-\beta)\overline{\Pi}_0)\right]$. Also note that $\overline{\Pi}_0=\delta\left[\alpha_0\overline{\Pi}_1+(1-\alpha_0)\overline{\Pi}_0\right]$. Therefore, $\Pi_0-\overline{\Pi}_0=-w_0+\delta(1-\alpha_0)\beta(\Pi_0-\overline{\Pi}_0)<\delta(1-\alpha_0)\beta(\Pi_0-\overline{\Pi}_0)$, where the inequality comes from $w_0>0$. Since $\delta\in(0,1)$ and $\beta\in(0,1]$, we discuss whether $\alpha_0=1$. If $\alpha_0<1$, the inequality cannot be satisfied, contradicting $w_0>0$. If $\alpha_0=1$, $\overline{\Pi}_0=\delta\overline{\Pi}_1$ and then $\Pi_0=-w_0+\delta\overline{\Pi}_1<\overline{\Pi}_0$, which contradicts (B-IC0) and also indicates that $w_0=0$. Therefore, $w_0=0$ and $\Pi_0=\overline{\Pi}_0$. Thus both (B-IC1) and (B-IC0) bind.

We now show that (P-IC1) is slack. Suppose it binds, so $U_1=\overline{U}$. Then given $w_0=0$ and $\delta<1$, plug in $U_1=\overline{U}$ and get $U_0=\delta\left[\alpha_0\overline{U}+(1-\alpha_0)(\beta U_0+(1-\beta)\overline{U})\right]<(1-\alpha_0)\beta U_0+(1-(1-\alpha_0)\beta)\overline{U}$. This inequality implies that $U_0<\overline{U}$, which contradicts (P-IC0). So (P-IC1) is slack. □

Lemma A.2. *For any buyer who directly contracts with producers, maximizing* Π_1 *is equivalent to minimizing* w_1.

Proof. Based on Lemma A.1, the buyer's continuation payoffs can be written as $\Pi_1=\frac{1-\delta(1-\alpha_0)}{1-\delta\alpha_0\alpha_1-\delta(2-\alpha_0-\alpha_1)+\delta^2(1-\alpha_0)(1-\alpha_1)}(y-w_1)$, and $\Pi_0=\frac{\delta\alpha_0}{1-\delta(1-\alpha_0)}\Pi_1$. Since $\frac{\partial\Pi_1}{\partial w_1}<0$, a buyer's problem is equivalent to minimize w_1 subject to producer's incentive constraints. □

A.4 Proof of Optimal Direct Contract, Lemma 2

For simplicity, we write w_1 as w in the rest of the proof. We complete the proof by analyzing and comparing the terms in the optimal bilateral relational contracts when choosing $\beta=0$ or $\beta>0$.

Choice 1: $\beta=0$. If the optimal choice of β is 0, $U_1=w-c+\delta\left[(1-\alpha_1)U_1+\alpha_1\overline{U}\right]$. The buyer optimally chooses w subject to a binding (P-IC-e), which gives $w_{DS}=\frac{1}{\delta(1-\alpha_1)}c+(1-\delta)\overline{U}$.

Choice 2: $\beta\in(0,1]$. If the optimal choice of β is greater than 0, a buyer's optimization problem becomes $\min_{w,\beta} w$ subject to (P-IC-e), (P-IC0), (1), (2), $0<\beta\le 1$, and $w\ge 0$.

The Lagrangian is given by

$$
\begin{aligned}
L = w &+ \lambda_1(U_1 - (w - c + \delta((1-\alpha_1)\cdot U_1 + \alpha_1(\beta U_0 + (1-\beta)\overline{U}))))\\
&+ \lambda_2(U_0 - \delta(\alpha_0 \cdot U_1 + (1-\alpha_0)\cdot(\beta U_0 + (1-\beta)\overline{U})))\\
&+ \mu_1(w + \delta\overline{U} - U_1) + \mu_2(\overline{U} - U_0) + \mu_3(\beta - 1) + \mu_4(-w).
\end{aligned}
$$

The Kuhn-Tucker conditions are given by $\frac{\partial L}{\partial w} = 1 + \mu_1(1 - \frac{\partial U_1}{\partial w}) - \mu_2\frac{\partial U_0}{\partial w} - \mu_4 \leq 0$; $\frac{\partial L}{\partial w}w = 0$; $\frac{\partial L}{\partial \beta} = -\mu_1\frac{\partial U_1}{\partial \beta} - \mu_2\frac{\partial U_0}{\partial \beta} - \mu_3 = 0$; $\lambda_1, \lambda_2 > 0$ $\mu_1, \mu_2, \mu_3, \mu_4 \geq 0$; $\mu_1(w + \delta\overline{U} - U_1) = 0$; $\mu_2(\overline{U} - U_0) = 0$; $\mu_3(1-\beta) = 0$; and $\mu_4 w = 0$.

By taking derivatives on both sides of equations (1) and (2) with respect to w, we get

$$\frac{\partial U_1}{\partial w} = 1 + \delta\Big[(1-\alpha_1)\frac{\partial U_1}{\partial w} + \alpha_1\beta\frac{\partial U_0}{\partial w}\Big], \tag{A1}$$

$$\frac{\partial U_0}{\partial w} = \delta\Big[\alpha_0\frac{\partial U_1}{\partial w} + (1-\alpha_0)\beta\frac{\partial U_0}{\partial w}\Big]. \tag{A2}$$

By taking derivatives on both sides of equations (1) and (2) with respect to β, we get

$$\frac{\partial U_1}{\partial \beta} = \delta\Big[(1-\alpha_1)\frac{\partial U_1}{\partial \beta} + \alpha_1(U_0 - \overline{U} + \beta\frac{\partial U_0}{\partial \beta})\Big], \tag{A3}$$

$$\frac{\partial U_0}{\partial \beta} = \delta\Big[\alpha_0\frac{\partial U_1}{\partial \beta} + (1-\alpha_0)(U_0 - \overline{U} + \beta\frac{\partial U_0}{\partial \beta})\Big]. \tag{A4}$$

We proceed by the following steps.

Step 1: Show that $\mu_4 = 0$ and $w > 0$. Suppose that $w = 0$. Then

$$U_1 = -c + \delta\Big[(1-\alpha_1)U_1 + \alpha_1(\beta U_0 + (1-\beta)\overline{U})\Big] < (1-\alpha_1)U_1 + \alpha_1(\beta U_0 + (1-\beta)\overline{U}),$$

where the inequality comes from that $c > 0$ and $\delta < 1$. The inequality above implies that $U_1 < \beta U_0 + (1-\beta)\overline{U} < U_0$, where the second inequality comes from (P-IC0). Meanwhile,

$$U_0 = \delta\Big[\alpha_0 U_1 + (1-\alpha_0)(\beta U_0 + (1-\beta)\overline{U})\Big] < \alpha_0 U_1 + (1-\alpha_0)(\beta U_0 + (1-\beta)\overline{U}) < U_0,$$

where the first inequality comes from $\delta < 1$ and the second inequality comes from (P-IC0) and $U_1 < U_0$. Since $U_0 < U_0$ can never be true, we know that $w > 0$ and thus $\mu_4 = 0$.

The implication for $w > 0$ is that $\frac{\partial L}{\partial w} = 1 + \mu_1(1 - \frac{\partial U_1}{\partial w}) - \mu_2 \frac{\partial U_0}{\partial w} = 0$.

Step 2: Discuss the values of μ_1 and μ_2. **Case 1)** $\mu_1 = \mu_2 = 0$. In this case, $\frac{\partial L}{\partial w} = 0$ is violated, indicating that this case is not possible. In other words, at least one of two incentive compatibility constraints bind.

Case 2) $\mu_1 = 0$ and $\mu_2 > 0$. In this case, $U_1 > w + \delta\overline{U}$ and $U_0 = \overline{U}$. Solve U_1 and w based on equation (1), and we get $U_1 = \frac{1-\delta(1-\alpha_0)}{\delta\alpha_0}\overline{U}$, and $w = c + \frac{(1-\delta)^2 - \delta(1-\delta)(\alpha_0+\alpha_1)}{\delta\alpha_0}\overline{U}$. So $\frac{\partial U_1}{\partial \beta} = \frac{\partial U_0}{\partial \beta} = 0$. Meanwhile, $\frac{\partial L}{\partial w} = 0$ and $\mu_1 = 0$ imply that $1 = \mu_2 \frac{\partial U_0}{\partial w}$. $\frac{\partial L}{\partial \beta} = 0$ implies $\mu_3 = 0$, namely $\beta < 1$.

There are two conditions that need to be satisfied for this case to be feasible and optimal. First, for feasibility, the solved w and U_1 need to satisfy $U_1 > w + \delta\overline{U}$. After plugging in U_1 and w as functions of $\overline{U}$, this requires that $\frac{(1-\delta)\overline{U}}{c} > \frac{\alpha_0}{1-\alpha_1}$. Second, since we are considering the case where choosing $\beta > 0$ is weakly better than choosing $\beta = 0$, the solved w needs to be lower than w_{DS}, which gives $\frac{(1-\delta)\overline{U}}{c} \leq \frac{\alpha_0}{1-\alpha_1}$. These two conditions contract each other, indicating that this case is not possible.

Case 3) $\mu_1 > 0$ and $\mu_2 > 0$. In this case, both incentive compatibility constraints bind, which give $U_1 = w + \delta\overline{U}$ and $U_0 = \overline{U}$. Under binding (P-IC-e) and (P-IC0), equations (1) and (2) can be satisfied only if $\frac{(1-\delta)\overline{U}}{c} = \frac{\alpha_0}{1-\alpha_1}$. In that case, the solved w also coincides with w_{DS}, and a buyer is thus indifferent among choosing any value of $\beta \in [0, 1]$.

Case 4) $\mu_1 > 0$ and $\mu_2 = 0$. In this case, $U_1 = w + \delta\overline{U}$, $U_0 > \overline{U}$, $\frac{\partial L}{\partial w} = 1 + \mu_1(1 - \frac{\partial U_1}{\partial w}) = 0$, and $\frac{\partial L}{\partial \beta} = -(\mu_1 \frac{\partial U_1}{\partial \beta} + \mu_3) = 0$. We discuss whether $\beta = 1$ or not. If $\beta \neq 1$, $\mu_3 = 0$, then $\frac{\partial U_1}{\partial \beta} = 0$. Given that, equation (A3) suggests that $\frac{\partial U_0}{\partial \beta} < 0$, while equation (A4) suggests that

$\frac{\partial U_0}{\partial \beta} > 0$. Therefore, a contradiction exists, indicating that $\beta = 1$. Given $\beta = 1$ and $\mu_3 > 0$, $w_{DI} = \left(\frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_0}{(1-\alpha_1)+\frac{\delta}{1-\delta}\alpha_0}\right)c + \left(\frac{1+\frac{\delta}{1-\delta}\alpha_0}{(1-\alpha_1)+\frac{\delta}{1-\delta}\alpha_0} - \delta\right)\overline{U}$.

In sum, if a buyer decides to choose $\beta > 0$, she will optimally choose $\beta = 1$ with paying w_{DI} when her demand is 1. The comparison between choice 1 and choice 2 hinge on comparing w_{DS} and w_{DI}. It turns out that $w_{DS} < w_{DI}$ if and only if $\frac{(1-\delta)\overline{U}}{c} > \frac{\alpha_0}{1-\alpha_1}$. Therefore, whenever the condition above holds, the buyer chooses $\beta = 0$ (separating from the producer when demand becomes 0) and pays w_{DS} when her demand is 1. Otherwise, she chooses $\beta = 1$ (retaining the producer when demand becomes 0) and pays w_{DI} when her demand is 1.

A.5 Proof of Optimal Managed Contract, Lemma 3

Observe that the continuation payoffs and incentive compatibility constraints for a manager in a managed contract are similar with those for a producer in a direct contract. The only differences are that the "cost of production" for a manager is w_M and its continuation value after separating from a buyer, given by $\overline{V}$, is 0. Since $\frac{(1-\delta)\overline{V}}{w_M} \leq \frac{\alpha_0}{1-\alpha_1}$ for any α_0 and α_1. Therefore, the value of p is determined by replacing c with w_M and $\overline{U}$ with 0 in w_{DI}.

A.6 Proof of Optimal Contractual Choice, Proposition 1

Step 1: Compare direct contract with separation and direct contract with idleness. From Lemma 2, it is immediate that if $\overline{U} > \overline{U}^{ei} \equiv \frac{c}{1-\delta}\alpha_0$, there exists a unique cutoff $\overline{\alpha}_1^{ei} \in (0, 1)$ such that $w_{DS} < w_{DI}$ if and only if $\alpha_1 < \overline{\alpha}_1^{ei}$.

Step 2: Compare direct contract with separation with managed contract. Observe that $w_{DS} < p$ if and only if $f(\alpha_1) \equiv \frac{1}{1-\alpha_1}\frac{c}{\delta} + (1-\delta)\overline{U} - \frac{1}{\delta - \frac{\delta}{1+\frac{\delta}{1-\delta}\alpha_0}\alpha_1}(\frac{c}{\delta} + (1-\delta)\overline{U}) < 0$. Observe that, when $\alpha_0 = 0$, $f(\alpha_1)\big|_{\alpha_0=0} = (1 - 1/\delta)\frac{c}{\delta}\frac{1}{1-\alpha_1} + (1 - \frac{1}{\delta(1-\alpha_1)})(1-\delta)\overline{U} < 0$. In this case, $w_{DS} < p$ for sure, so $\overline{\alpha}_1^{eo} = 1$. When $\alpha_0 > 0$, observe that $f(0) = (1 - \frac{1}{\delta})\kappa < 0$, and $f(1)$ goes to infinity. We now show that the continuous function $f(\alpha_1)$ intersects with 0 only once. Note that $\frac{\partial f(\alpha_1)}{\partial \alpha_1} = \frac{\zeta(\zeta\frac{c}{\delta}-\kappa)\alpha_1^2 - 2(c-\kappa)\zeta\alpha_1 + (\delta c - \zeta\kappa)}{(1-\alpha_1)^2(\delta-\zeta\alpha_1)^2}$, where $\zeta = \frac{\delta}{1+\frac{\delta}{1-\delta}\alpha_0}$ and

$\kappa = \frac{c}{\delta} + (1-\delta)\overline{U}$. Observe that the numerator is a quadratic equation, where the coefficient of α_1^2 is negative, the coefficient of α_1 is positive, and the constant $\delta c - \zeta(\frac{c}{\delta} + (1-\delta)\overline{U})$ can be positive or negative. Therefore, $f(\alpha_1)$ is either strictly increasing, or is first decreasing then increasing. In either case, $f(\alpha_1)$ intersects with 0, with the intersect being $\overline{\alpha}_1^{eo} \in (0,1)$. In sum, there exists a unique cutoff $\overline{\alpha}_1^{eo} \in (0,1]$ such that $w_{DS} < p$ if and only if $\alpha_1 < \overline{\alpha}^{eo}$.

Step 3: Compare direct contract with idleness and managed contract. Observe that $w_{DI} < p$ if and only if $g(\alpha_1) \equiv \frac{1+\frac{\delta}{1-\delta}\alpha_0}{1+\frac{\delta}{1-\delta}\alpha_0 - \alpha_1}(\frac{2\delta-1}{\delta}\overline{U} - \frac{1-\delta}{\delta}\frac{c}{\delta}) < \delta\overline{U}$. Observe that $g(0) = \frac{2\delta-1}{\delta}\overline{U} - \frac{1-\delta}{\delta}\frac{c}{\delta}$ and $g(1) = \frac{1+\frac{\delta}{1-\delta}\alpha_0}{\frac{\delta}{1-\delta}\alpha_0}(\frac{2\delta-1}{\delta}\overline{U} - \frac{1-\delta}{\delta}\frac{c}{\delta})$. Also observe that $g(0) - \delta\overline{U} = \frac{-(1-\delta)^2}{\delta}\overline{U} - \frac{1-\delta}{\delta}\frac{c}{\delta} < 0$. If $\overline{U} < \overline{U}^{io} \equiv \frac{(1-\delta(1-\alpha_0))c}{\delta(\delta(2-(1-\delta)\alpha_0)-1)}$,g(1) $< \delta\overline{U}$ and thus the inequality holds for sure. Otherwise, since $g(\alpha_1)$ is strictly increasing in α_1, by the intermediate value theorem, there exists a unique cutoff $\overline{\alpha}_1^{io} \in (0,1)$ such that $w_{DI} < p$ if and only if $\alpha_1 < \overline{\alpha}_1^{io}$. Otherwise, $w_{DI} < p$ for sure.

Step 4: Put it together. Let $\overline{U}^* = \overline{U}^{io}$ and $\alpha^* = \min\{\overline{\alpha}_1^{io}, \overline{\alpha}_1^{eo}\}$. If $\overline{U} < \overline{U}^{io}$, $w_{DI} < p$ by Step 3. Otherwise, if $\overline{U} \geq \overline{U}^{io}$ there are two cases. If $w_{DS} < w_{DI}$, managed contract is optimal when $p < w_{DS}$, namely when $\alpha_1 > \overline{\alpha}_1^{eo}$. If $w_{DS} > w_{DI}$ managed contract is optimal when $p < w_{DI}$, namely when $\alpha_1 > \overline{\alpha}_1^{io}$.

A.7 Proof of Cross-sectional Comparisons, Corollary 1

First, note that $w_D(\alpha_i) > w_M$ for all $i \in \mathcal{I}_{DS} \cup \mathcal{I}_{DI}$. Therefore, $E[w_D(\alpha_i) \mid i \in \mathcal{I}_{DS} \cup \mathcal{I}_{DI}] > E[w_M | i \in \mathcal{I}_M]$. Second, note that $w_D(\alpha_i)$ takes on different values depending on α_i, which is heterogeneous across buyers, while w_M is constant. Therefore, $\text{Var}[w_D(\alpha_i) \mid i \in \mathcal{I}_{DS} \cup \mathcal{I}_{DI}] > \text{Var}[w_M \mid i \in \mathcal{I}_M] = 0$. Third, the separation rate is given by $\beta_i \alpha_{1i}$. Note that $\beta_i = 1$ for all $i \in \mathcal{I}_{DI}$, while $\beta_i = 0$ for all $i \in \mathcal{I}_{DS} \cup \mathcal{I}_M$. Fourth, the idleness for a buyer $i \in \mathcal{I}_{DI}$ is given by $(1-\pi_i)(1-\omega_i) > 0$. By contrast, for $i \in \mathcal{I}_{DS} \cup \mathcal{I}_M$, idleness is zero.

A.8 Proof of Managerial Impacts, Corollary 2

With the introduction of managers, the per-period payoff of buyer $i \in \mathcal{S}_0$ increases by $y - p(\alpha_i)$. The per-period payoff of buyer $i \in \mathcal{S}_I$ increases by $w_{DI}(\alpha_i) - p(\alpha_i)$. The per-period payoff of buyer $i \in \mathcal{S}_S$ increases by $w_{DS}(\alpha_i) - p(\alpha_i)$. The per-period payoff of the remaining buyers are unchanged. The average producer pay per unit effort falls, since $w_D(\alpha_i) > w_M$ for all $i \in \mathcal{S}_I \cup \mathcal{S}_S$. Producer separation rates and idleness also fall, since buyers switch from direct contracts with either idleness and separation to managed contracts in which buyers are never idle and never separate. The measure of buyers who receive services increases by $|\mathcal{S}_0|$. The total measure of services provided in the economy increases by $\int_{\mathcal{S}_0} \pi_i dF$. However, for $i \in \mathcal{S}_I$, the steady-state measure of producers matched to these buyers fall, since idleness falls. The total measure of matched producer $(n_M + n_B)$ therefore increases if and only if $\int_{\mathcal{S}_0} \pi_i dF > \int_{\mathcal{S}_I} (1 - \pi_i) dF$.

A.9 Proof of Equilibrium with Specialization, Proposition 3

Based on Lemma 2 and 3, the buyer's post-matching continuation payoffs when she has demand is $\Pi_D(\alpha_i) = \frac{y+\Delta_i}{1-\delta} - \left[\overline{U} + \frac{c}{(1-\alpha_i)\delta(1-\delta)}\right]$ if directly contracting with a specialist producer. It is $\Pi_G = \frac{y}{1-\delta} - \left[\overline{U} + \frac{c}{\delta(1-\delta)}\right]$, if directly contracting with a generalist producer. It is $\Pi_M(\alpha_i) = \frac{y+\Delta_i}{1-\delta} - \left[\frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_i}{1+\frac{\delta}{1-\delta}\alpha_i - \alpha_i}(\overline{U} + \frac{c}{\delta(1-\delta)})\right]$ if contracting with a manager.

Three observations follow. First, a buyer prefers to directly contract with a specialist than directly contract with a generalist if and only if $\Pi_B(\alpha_i) \geq \Pi_G$, or, $\Delta_i \geq \frac{\alpha_i}{1-\alpha_i}\frac{c}{\delta}$. Second, a buyer prefers to directly contract with specialists rather than contract with a manager if and only if $\left[\frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_i}{1+\frac{\delta}{1-\delta}\alpha_i - \alpha_i} - 1\right]\left[\overline{U} + \frac{c}{\delta(1-\delta)}\right] \geq \frac{\alpha_i}{1-\alpha_i}\frac{c}{\delta(1-\delta)}$. By Proposition 1, there exists a unique cutoff α^{ei} such that the buyer prefers to contract with a manager if and only if $\alpha_i > \alpha^{ei}$. Third, a buyer prefers a managed contract over direct contract with a generalist if and only if $\Delta_i \geq \overline{\Delta}(\alpha_i) \equiv \left[\frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_i}{1+\frac{\delta}{1-\delta}\alpha_i - \alpha_i} - 1\right]\left[\overline{U} + \frac{c}{\delta(1-\delta)}\right]$. Therefore, the buyer choose to contract with a manager if and only if α_i and Δ_i are both large enough.

A.10 Proof of Managerial Impacts with Specialization, Corollary 3

Let $\mathcal{S}_G$ denote buyers who switch from direct contracts with generalists to managed contracts with specialists when managers become available. Let $\mathcal{S}_B$ be the set of buyers who switch from direct contracts with specialists to managed contracts with specialists. Given the assumption on the distribution of (Δ_i, α_i), $|\mathcal{S}_G|, |\mathcal{S}_B| > 0$. The contractual choice of the remaining buyers are unchanged. Therefore, the measure of specialists unambiguously increase with managerial coordination.

The Bellman equation for unmatched producers can be rewritten as $\overline{U} = \frac{v}{u}\frac{E[v_i U_{1i}]}{v} + \left(1 - \frac{v}{u}\right)\delta\overline{U}$. With the introduction of managers, more producers are contracted as specialists under managed contracts without separation, so v and $E[v_i U_{1i}]$ both fall. This implies that $\frac{v}{u}$ increases. It follows that the measure of unmatched producers $n_N = u - v$ falls.

B Model with Manager Reputational Concerns

Our baseline model follows Shapiro and Stiglitz (1984) in assuming that producers are motivated to perform through the threat of contract termination. Another source of motivation, as modeled by Klein and Leffler (1981), is that producers may lose reputational capital when they renege on promises to deliver high-quality services. In this subsection, we study how manager reputation concerns shape the choice between managed and direct contracting. We then ask: How does the arrival of communication technologies like the Internet and social media affect the extent of managerial coordination and the division of labor?

We consider a multi-task economy with reputable managers, where low effort by a producer is communicated with some probability to other buyers who can withhold future business from the producer's matched manager. Specifically, we assume that with probability $\gamma \in [0, 1]$, the effort choice by a producer contracted by the manager is observed by another buyer, who is drawn among all buyers with uniform probability.[19]

[19]Here the manager will not wish to renege on more than one of its clients if she does not wish to do so for a single client, since a buyer may learn of bad service provided to multiple clients from word-of-mouth communication but can only punish maximally once.

The parameter γ measures the *ease of word-of-mouth communication* about manager performance as enabled by communication technologies. A manager's mean continuation value from being matched with a buyer is therefore $\tilde{V} = \frac{1}{|\mathcal{I}|}\int_{\mathcal{I}_O} \pi_i V_{1i} + (1-\pi_i)V_{0i} dF > 0$. When $\gamma > 0$, the manager's binding IC constraint becomes

$$V_1 \geq p_1 + \delta(-\gamma\tilde{V}). \tag{M-IC-w'}$$

As γ increases, the manager faces a harsher threat of multilateral punishment. Therefore, a reduced mark-up is needed to incentivize the manager to perform. Consequently, the unit cost of managed service decreases as the ease of communication increases. This in turn increases managerial coordination, and thereby enables specialization and reduces the measure of unmatched producers.

Proposition B.1. *A unique steady-state equilibrium exists in a multi-task economy with reputable managers. As the ease of word-of-mouth communication increases, the measure of managed producers increases, the measure of specialist producers increases, and the measure of unmatched producers falls.*

Proof. In a multi-task economy with reputable managers, the service fee is given by

$$p(\alpha_i) = \left(\frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_i}{(1-\alpha_i)+\frac{\delta}{1-\delta}\alpha_i}\right)w_M + \left(\frac{1+\frac{\delta}{1-\delta}\alpha_i}{(1-\alpha_i)+\frac{\delta}{1-\delta}\alpha_i} - \delta\right)(-\gamma\tilde{V})$$

This implies that

$$\Pi_M(\alpha_i) = \frac{y+\Delta_i}{1-\delta} - \left[\frac{1}{\delta}\frac{1+\frac{\delta}{1-\delta}\alpha_i}{1+\frac{\delta}{1-\delta}\alpha_i - \alpha_i}(\overline{U} + \frac{c}{\delta(1-\delta)})\right] + \left(\frac{1+\frac{\delta}{1-\delta}\alpha_i}{(1-\alpha_i)+\frac{\delta}{1-\delta}\alpha_i} - \delta\right)\frac{\gamma\tilde{V}}{1-\delta},$$

As γ increases, $\Pi_M(\alpha_i)$ rises, so buyers switch to managed contracts with specialists from direct contracts with both generalists and specialists. Therefore, by the same logic as Corollary 3, the measure of specialists increases, while the measure of unmatched producers decreases.

□

These predictions help make sense of recent macroeconomic trends. Bergeaud et al. (Forthcoming) provide causal evidence that the rise of broadband internet increased the employment share of professional service firms and the average occupational concentration of firms. Correlational evidence shows that as the employment share of professional service firms has increased, workers and firms have become increasingly specialized, and that unemployment has fallen (Katz and Krueger 1999; Song et al. 2019; Handwerker 2023).

C Model with Exogenous Producer Population

The main text assumed that the number of producers is determined via endogenous entry. In this appendix, we analyze an economy where the number of producers is fixed exogenously at $n > 1$. The main difference is that when the producer population is fixed, the presence of managers alters $\overline{U}$ by changing the overall demand for producers, and thereby alters equilibrium outcomes.

We first show that if n is sufficiently large, then the producer market becomes very slack, buyers prefer to enter contracts with idleness, and there is no managerial coordination.

Proposition C.1. *Assume the number of producers n is exogenously fixed. There exists a unique steady-state equilibrium. If n is sufficiently large, then there are no managed contracts.*

Proof. Given $\overline{U}$, we can compare the values of $w_{DS}(\alpha_i)$, $w_{DI}(\alpha_i)$, $p(\alpha_i)$, and y for each i using (6), (7), and (11) to determine $\mathcal{I}_{DS}$, $\mathcal{I}_{DI}$, and $\mathcal{I}_M$. Having derived these, we can obtain unique values for v_i, v, n_B, and n_M from (12), (13), (14), and (15). Plugging $w_{DS}(\alpha_i)$, $w_{DI}(\alpha_i)$, and $w_M(\alpha_i)$ into (18) yields a contraction mapping. Therefore a unique fixed point for $\overline{U}$ exists. By (16), (17), and (18), $\overline{U}$ is decreasing in n, and $\overline{U} \to 0$ as $n \to \infty$. The desired result follows by Proposition 1. □

We next compare economies with and without managers assuming that n is fixed and that some nonzero measure of buyers choose managerial coordination when possible. In this economy, introducing managers has both direct and indirect effects. The direct effect holds $\overline{U}$ fixed and was characterized in the previous subsection. Due to this effect, various types of buyers switch to managerial coordination, producer rents fall, and the number of matched producers may or may not increase. Without free entry, however, $\overline{U}$ also adjusts. A change in $\overline{U}$ affects all of the endogenous variables in the model, so managerial coordination has indirect spillover effects onto all producers.

The sign of the resulting change in $\overline{U}$ depends on the distribution of buyer types α_i. If a large set of buyers switch from being unmatched to matched, then $\overline{U}$ would increase due to reduced wait time in a tightened producer market. This indirect effect raises the pay of all producers. On the other hand, if no buyers switch from being unmatched to being matched, then $\overline{U}$ would unambiguously fall due to producer rent reductions associated with initially matched buyers switching to managerial coordination. This indirect effect instead reduces the pay of all producers.